

Khushal Pareta

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Profile

Data science professional with hands-on experience building end-to-end AI/ML solutions, from scalable data pipelines to model deployment and monitoring. Proven ability to deliver production-grade Retrieval-Augmented Generation (RAG) systems using LangChain, Hugging Face models, and vector databases. Strong background in database performance tuning (PostgreSQL, connection pooling), ETL orchestration (PySpark, Airflow), MLOps (MLflow, Docker, ELK), and cloud platforms (AWS, Azure). Combines deep technical skills with the ability to translate complex data into actionable insights for national-scale projects.

Core Competencies

Machine Learning & AI: Python, TensorFlow/Keras, Scikit-learn, Statistical Modelling, Feature Engineering, Experiment Tracking (MLflow), LLM Fine-tuning
Generative AI & LLM Apps: LangChain, Hugging Face Transformers, Prompt Engineering, RAG Pipelines, Vector Databases (Chroma/FAISS), Semantic Search
Data Engineering: PySpark, SQL, Hadoop, Hive, Kafka, Apache Airflow, ETL/ELT, Data Lake/Warehouse Concepts
MLOps & DevOps: Docker, Docker Compose, Git, CI/CD, ELK Stack (Elasticsearch, Logstash, Kibana), Linux Administration, Cron/pg_cron
Databases & Storage: PostgreSQL (Query Tuning, PL/pgSQL, pgBouncer, dblink), MySQL, MongoDB, AWS S3, Azure Blob
Cloud & Monitoring: AWS (Lambda, Rekognition, SNS), Azure (Event Hubs, ML Studio), Prometheus/Grafana (basics), Power BI, Tableau, Streamlit
Data Analytics & BI: Power BI, Tableau, Advanced Excel, Statistical Analysis, Data Visualization, KPI Dashboards, Demand Forecasting, Inventory Analytics

Professional Experience

Data Analyst

Laxmi Traders, Aklera, Rajasthan

July 2022 – August 2024

- Developed **Power BI** dashboards and automated MIS reports to monitor procurement, inventory levels, warehouse stock movement, and daily sales across multiple grain commodities, significantly reducing manual reporting effort.
- Designed Python- and SQL-based **ETL pipelines** to consolidate procurement, sales, and inventory data into a centralized PostgreSQL database, improving reporting accuracy and data availability.
- Built statistical models for **commodity price trend analysis** and demand forecasting using historical market and sales data, supporting procurement planning and inventory optimization.
- Created supplier and procurement analytics dashboards to evaluate purchase prices, delivery performance, and procurement volumes, enabling data-driven sourcing decisions and improving procurement efficiency.
- Automated inventory reconciliation, profit analysis, and data validation workflows using Python and SQL, significantly reducing manual errors and accelerating month-end business reporting.

Project Engineer (AI/ML & Data Pipelines)

C-DAC, Juhu, Mumbai

March 2025 – Present

- Architected and deployed an **LLM-powered conversational assistant** for the Maritime Knowledge Cluster (Govt. of India) using **LangChain**, Hugging Face models, and a vector database; implemented RAG with semantic search and prompt engineering, serving 100+ maritime stakeholders.
- Engineered PySpark ETL pipelines for the **National Family Health Survey (NFHS)**, reducing data processing latency by 40% through query optimization, indexing, and PostgreSQL connection pooling (pgBouncer).
- Built a Python-based **error classification model** for real-time vessel tracking (LRIT), cutting false-positive ingestion alerts by 25%; deployed the **ELK Stack** for centralized logging and dashboards.
- Optimised **PostgreSQL** production servers for the Cotton Procurement System (COTBIZ) – query tuning, dblink cross-database queries, and pgBouncer – achieving 30% higher transaction throughput; automated backup/alerting with cron and pg_cron.
- Containerized analytics micro-services and ML pipelines with **Docker** and Docker Compose; managed model lifecycle with **MLflow** (tracking, registry) for TensorFlow/Keras models.

Key Projects

Maritime Knowledge Cluster (MKC) – AI Assistant–Designed a RAG pipeline integrating LangChain, Hugging Face embeddings, and a vector store to ground LLM responses in official maritime regulations and circulars.

–Developed document ingestion & chunking workflows, API integration, and deployed the service on Linux servers using Docker.

–Applied prompt engineering and hyperparameter tuning to improve answer relevance and latency.

Bharat Trace – Supply Chain Traceability–Administered PostgreSQL databases with connection pooling (pgBouncer), query optimisation, and automated ETL pipelines for end-to-end supply chain visibility.

–Configured ELK Stack for monitoring data ingestion and application logs, reducing mean-time-to-detect issues.

–Containerised data services using Docker Compose and managed Linux-based deployments.

LRIT – Long Range Identification & Tracking–Implemented real-time vessel tracking data pipelines, integrating AIS data sources with secure communication layers.

–Developed Python sanity-check modules that classified data anomalies, improving system reliability.

–Built Kibana dashboards atop Elasticsearch for operational visibility of maritime traffic.

NFHS – National Family Health Survey–Created PySpark-based ETL pipelines and validation workflows to process nationwide survey data for the Ministry of Health.

–Designed interactive reporting dashboards (Streamlit, Power BI) that communicated key health indicators to policy makers.

–Tuned PostgreSQL indexing and query performance, accelerating analytical queries by over 35%.

CCI-CPPS – Cotton Procurement Platform–Developed a digital procurement platform backend with Python and PostgreSQL, ensuring traceability and transparency in cotton trading.

–Automated data validation, report generation, and user notification workflows, reducing manual reconciliation efforts.

Education

Post Graduate Diploma in Big Data Analytics

2024–2025

Centre for Development of Advanced Computing (CDAC), Mumbai

Bachelor of Technology in Computer Science

2018–2022

Poornima College of Engineering, Jaipur

Achievements

– Awarded **1st Rank** for Academic Excellence in CDAC PG-DBDA (September 2024).